

Recent developments of MPPT techniques for PV systems under partial shading conditions: a critical review and performance evaluation

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Abstract: This review covers global maximum power point tracking (GMPPT) methods for photovoltaic (PV) systems under partial shading conditions. Unlike the previous review works that primarily focused on soft computing and hybrid GMPPT, this study gives exclusive attention to the improvement achieved by the conventional MPPT (perturb and observe, hill climbing, and incremental conductance). The improved methods include the popular $0.8 \times V_{oc}$ model and, more recently, the skipping algorithms. In addition to providing qualitative descriptions of the available techniques, this work also attempts to provide a fair evaluation of GMPPT to determine their comparative performances. The competing algorithms, which are selected to represent every category (conventional and soft computing and hybrid MPPT), are benchmarked under carefully selected operating conditions and shading scenarios. The evaluation is focused on four main criteria: tracking accuracy, convergence time, length of voltage fluctuations, and transient efficiency during the search for the global maximum power point. The results obtained from this study can become a basis for researchers and designers to select the best MPPT technique for their respective applications.

1. Introduction

In the last decade or so, maximum power point tracking (MPPT) methods that are used to extract the maximum possible energy from photovoltaic (PV) systems have vastly proliferated. This trend is understandable due to the rapid expansion in the number of installations that need to maximise the PV energy yield. One of the issues that is of concern for MPPT designers is the ability of the algorithm to track the global maximum power point (GMPP) during the occurrence of partial shading. Under this condition, the MPPT is no longer tracking a single-peak $P-V$ curve; the problem becomes much more complex as the curve now exhibits multiple power peaks with unpredictable trajectories and shapes. The MPPT that addresses this problem is often known as the global MPPT (GMPPT) method.

Conventional MPPTs, such as perturb and observe (P&O), hill climbing, and incremental conductance (InCond), are known to be most efficient for uniform irradiance, where the peak is unique. However, they are not amenable when the system is subjected to partial shading conditions. Often, the algorithm is trapped at a local peak, resulting in power loss [1–5]. Thus, modification to the conventional MPPT is needed to ensure that the GMPP is consistently tracked. The most popular method is based on the observation that the peaks (local and global) are located in the vicinity of $0.8 \times V_{oc}$ multiples (i.e. $0.8 \times V_{oc}$, $2 \times 0.8 \times V_{oc}$, $3 \times 0.8 \times V_{oc}$, ...), where V_{oc} is the open-circuit voltage of the PV modules. This concept, which is widely referred to as the $0.8 \times V_{oc}$ method, was initially proposed by Patel and Agarwal [1]. Since then, the idea has inspired numerous permutations that allow the conventional MPPT to handle partial shading effectively. More recently, another category of solutions, called the skipping mechanism, was introduced to improve the conventional MPPT [6,

7]. The underlying idea is to discard specific voltage intervals within the $P-V$ curve, in which the GMPP is guaranteed to be absent. The area to be scanned is now reduced, thus making rapid GMPP tracking possible. On the basis of this concept, several advanced skipping schemes are successfully implemented for the various proposed conventional MPPTs. Examples of such works can be found in [8–10].

Apart from the conventional MPPT, the soft computing (SC)-based GMPPT is also gaining popularity to mitigate the partial shading problem [11–14]. With the MPPT assigned as an optimisation problem, a metaheuristic search can be feasibly used to locate the GMPP. A large variety of SC techniques are adopted. Some utilise the popular methods, namely, particle swarm optimisation (PSO), differential evolution (DE), cuckoo search (CS), genetic algorithms (GAs), ant colony optimisation (ACO), and artificial bee colony (ABC). Others experimented on lesser-known or recent methods such as bat search algorithm (BSA), grey wolf optimisation (GWO), firefly algorithm (FFA), and fish swarm algorithm (FSA). Hybrid approaches, where two or more GMPPT methods are combined to exploit each other's advantages, have received growing interest. For instance, Seyedmahmoudian *et al.* [15] proposed a combination of two SC techniques (PSO with DE), while in [16], the PSO is combined with the skip-judge method.

The body of literature concerning partial shading is extensive; therefore, many review papers were published on this subject [16–18]. Typically, a review summarises the status, progress, and future outlook of various GMPPT techniques. More importantly, a review paper is always looked upon as a convenient one-stop information centre for a broad spectrum of issues that have helped shape the advancement and direction of the technology. Needless to say, it is also an important source of reading material for new researchers venturing in this field. Table 1 summarises the latest and

Table 1 Summary of the latest (recent five years) review papers on GMPPT under partial shading condition

Author, Year	GMPPT techniques being reviewed	Comments
Rezk <i>et. al.</i> [19], 2019	category: SC: PSO, DE, FPA, TLBO, GWO, WCA, HSA, DFA, MBA, ALO, ICA, RMO, MVO, WOA, SCA, GSA, CS, FA, GA, JA	• twenty GMPPT methods are investigated • all algorithms are tested under the same platform, i.e. PV-boost feeding a battery. • the shading patterns and sampling time are standardised • statistical analysis is carried out using the following parameters: relative error, root mean square error, mean absolute error, standard deviation, successful rate, and efficiency • a critical evaluation is partially performed. Several well-known GMPPT approaches, particularly hybrid methods and improved conventional based on the $0.8V_{OC}$ model and SM are not included in the analysis. • the optimiser parameters of the SC methods under study are not discussed (it is unknown whether the parameters used are optimal or not)
Eltamaly <i>et. al.</i> [14], 2019	category: SC: SO, GWO, FPA, ACO, ABC, FA, TLBO, SA category: hybrid methods: PSO-FLC, GWO-FLC GWO-P&O, GWO-FLC-PSO, PSO-IC, PSO-P&O, SA-PSO, SA-IC, SA-P&O, ABC-P&O	• twenty algorithms are reviewed • several important GMPPT approaches, particularly the improved conventional based on the $0.8V_{OC}$ model and SM are not included in the analysis • the evaluation made is not accurate. Only qualitative judgments are given, based on the results provided by the reviewed papers. • no critical evaluation of the performance of the reviewed methods is carried out
Belhachat and Larbes [20], 2019	category: modified conventional methods: IVCA, CSR, MPT, PINC, TSSBT, ES, EB, HSM, EA-P&O, AV, BDS, CRT category: SC: ICSO, MCSO, IGSA, IDE, MPV-PSO, modified FA, deterministic CS, MFO, A2MGO, modified AK-WO, MSSA category: hybrid methods: GWO-Beta, modified GA-FA, SSHC-ABC, GWO-GSO, PSO-OCC, ANN-SMC, OD-PSO, improved $0.8V_{OC}$ model and P&O, HTGA, ANN-PSO, ABC-P&O	• thirty-four GMPPT methods are reviewed • the methods are classified into four groups: optimisation-based, hybrid, PV characteristics-based, and others • many recent GMPPT, particularly the methods based on the $0.8V_{OC}$ model and SM are not included • the reviewed GMPPT are not evaluated under identical test conditions • since the results of the reviewed methods are directly collected from the original publications, they are not reasonably assessed
Belhachat and Larbes [21], 2018	category: modified conventional methods: modified Beta, modified FS, ESC, SCT, CF, ABSC, OC2S, RST, SCS, VPS, distributed PLC, OCC category: SC: PSO, ABC, ACO, SA, BA, FA, FWA, GWO, FPA, TLBO, MBA, S-Jaya, GSO, WOA, and HPO. category: hybrid methods: ANFIS-PI, INC-FFA, SA-PSO, FWA-P&O, Jaya-DE, P&O-ANN, scanning procedure and ANN, AFL-ANN, DANC-VT, MPF-IP&O, LI-PSO, PSO-P&O, IPSO-BELBIC, PSO-SFLA, GWO-P&O, PSOEM-FSA, WO-DE, modified P&O and checking algorithm, modified BA-P&O	• forty-seven GMPPT approaches are reviewed • the reviewed methods are classified into four groups: one-optimisation-based, mathematical model-based, hybrid, and other GMPPT techniques • many recent GMPPT (based on the $0.8V_{OC}$ model and SM) are not included • the comparative evaluation is not accurate since the parameter values (efficiency and tracking speed) are directly collected from original publications. However, the algorithms—in the original papers—are investigated under different test conditions
Li <i>et. al.</i> [22], 2018	category: SC: classified into two categories: (i) evolutionary algorithm: GA, EP, DE (ii) SI: PSO, ACO, CS, ABC, FA	• eight SC methods are reviewed • they are classified into two categories: evolutionary algorithms (EA) and SI-based methods • several other efficient SC approaches, methods based on the $0.8V_{OC}$ model and SM-based, and hybrid approaches are not included • the judgment for some criteria (tracking time and efficiency) is insufficient since it is qualitatively made according to the results published in the original works, which cannot lead to an accurate evaluation • no critical evaluation is made under identical test conditions

comprehensive GMPPT review papers published in the last five years. As can be observed, the focus of previous work is on SC and hybrid MPPT approaches. Clearly, less attention is given to developments in conventional GMPPT methods, i.e. $0.8 \times V_{OC}$ and the skipping mechanisms. Furthermore, a thorough study of the

literature reveals that none has provided a fair evaluation to determine the comparative performances of various GMPPT approaches. This condition is most likely due to the fact that each published algorithm is tested using a proprietary test bench and under specific operating conditions.

Author, Year	GMPPT techniques being reviewed	Comments
Seyedmahmoudian <i>et al.</i> [23], 2016	category: SC: ANN, FLC, PSO, ACO, GA, DE, RMO, CS, GWO, FA.	• eight SC-based methods are reviewed • only SC algorithms are considered, while GMPPT methods from other categories are omitted • an attempt to give analytical comparison based on several criteria, such as periodic tuning, simplicity, efficiency, and convergence speed • the comparison is qualitatively made according to the published data in the reviewed works using loosely words, such as poor, average, high • no consistent evaluation under identical test conditions is carried out. Thus, the judgment given is not accurate.

ABSC, Active bypassing of the shaded cells; AFL, augmented state feedback precise linearisation; A2MGO, artificial mountain ape optimisation; AV, artificial vision algorithm; ALO, antlion optimiser; AK, artificial killer; BA, bat search algorithm; BS, binary search; BI, bio-inspired algorithm; BELBIC, brain emotional learning-based intelligent controller; BDS, bypass diode scanning approach; CRT, cells reconfiguration technique; CT, conventional technique; CF, curve fitting; CSR, current source region detection of the I-V curve; DANC-VT, direct adaptive neural control and voltage traverse; DFA, dragonfly algorithm; EA-P&O, enhanced adaptive P&O; ES, enhanced scanning; EB, enhanced Bayesian; EA, evolutionary algorithms; EP, evolutionary programming; ESC, extremum seeking control; FS, Fibonacci search; GSA, gravitational search algorithm; HPO, human psychology optimisation; HTGA, hybrid Taguchi GA; HSA, harmony search algorithm; IVCA, $I-V$ curve approximation; HSM, high-speed MPPT module; ICSO, improved chicken swarm optimisation; Glowworm swarm optimization (GSO), glowworm swarm optimisation/golden-section optimisation; IGSA, improved gravitational search algorithm; IDE, improved DE; ICA, imperialist competitive algorithm; MBA, mine blast algorithm; MSSA, memetic slap swarm algorithm; MVO, multi-verse optimiser; MPV, modified particle velocity; MFO, moth-flame optimisation algorithm; MCSO, modified cat swarm optimisation; MPF-IP&O, model-based and improved P&O; OD, overall distribution; OC2S, open circuit two stages method; PSOEM-FSA, PSO optimised FSA with extended memory; PLC, programmable logic controller; PINC, power-increment algorithm; RMO, radial movement optimisation; RST, recurrent scan and track; SMC, sequential Monte Carlo; SCT, series capacitor transient; SSHC, single current sensor hill climbing; SCA, sine cosine algorithm; SM, skipping method; SCS, stepped comparison search; TLBO, teaching learning-based optimisation technique; TSSBT, two-stage searching and bisection techniques; VPS, various pre-planned segments estimation; WCA, water cycle algorithm.

On the basis of these discussions, this work is conducted to fulfil two main research gaps. First, the approaches to improve the conventional GMPPT, namely, $0.8 \times V_{oc}$ and the skipping techniques, will be treated more exclusively. Thus far, no review has been conducted on these methods; in a way, their inclusion is meant to complement the extensive reviews received by the SC and hybrid-based methods. Second, an attempt to provide a fair evaluation of GMPPT performance is carried out. The competing algorithms, which are selected to represent every category (conventional and SC and hybrid), are benchmarked under carefully selected operating conditions. These conditions include the same test system, simulation conditions, and shading scenarios. The results obtained from this study can become a basis for researchers and designers to select the best MPPT technique for their respective applications. An important detail to note is that this work will not cover the hardware approach to mitigate partial shadings, such as a microinverter, DC power optimiser, and PV array reconfigurations [24]. Despite their upcoming popularity, they are not grouped under GMPPT solutions because their concern is not on the tracking algorithms.

The remainder of this paper is organised as follows: in Section 2, a review of recently published GMPPT works is presented. They are categorised into four main groups: $0.8 \times V_{oc}$, skipping mechanism, SC, and hybrid approaches. In Section 3, a critical evaluation of selected GMPPT techniques from each of the categories mentioned above is carried out under the same test conditions. In Section 4, the main findings of the assessment are summarised. The performance of the competing algorithms is determined based on their ability to consistently track the GMPP, convergence time, voltage track, and transient efficiency. In the last section, discussions about future trends in the MPPT area and useful recommendations for researchers and workers in this field are given.

2 Review of recent GMPPT

The reviewed GMPPT methods are classified into four main categories: the modified conventional MPPT, the skipping mechanisms, SC algorithms, and hybrid approaches. SC and hybrid were extensively reviewed previously and are thus treated superficially here. The conventional MPPT and the skipping mechanisms are discussed in detail.

2.1 Modification of the conventional MPPT

2.1.1 The $0.8 \times V_{oc}$ method: Although the conventional MPPT techniques are very efficient under uniform insolation, they usually do not work well when subjected to partial shading [1–8]. To correctly track the GMPP, the algorithm needs to incorporate additional intelligence so that clear discrimination can be made between the local maximum power peaks (LMPPs) and the GMPP. One of the earliest ideas to improve the generic P&O is the so-called $0.8 \times V_{oc}$ method, proposed by Patel and Agarwal [1]. Using extensive data correlation and practical observations, the authors discovered several critical characteristics of the $P-V$ curve during partial shading

- Observation 1: all power peaks (local and global) always occur in the vicinity of the integral multiples of 80% of V_{oc_mod} ($0.8 \times V_{oc_mod}$). Thus, scanning only the areas that are near these points is sufficient.
- Observation 2: the peaks are displaced from each other by the multiple of V_{oc_mod} . This observation implies that the minimum displacement between two consecutive peaks (denoted by the variable ΔV_{GSTEP}) must also be $0.8 \times V_{oc_mod}$.
- Observation 3: the magnitude of the peaks increases when the operating point is traversed from either side of the $P-V$ curve. After the GMPP is reached, the magnitude of the subsequent peaks (if they are present) continuously decreases.

Inspired by this concept, Tey and Mekhilef [2] proposed a modified Incremental Conductance (IncCon) MPPT, which can rapidly locate the GMPP under partial shading condition. The idea is to simplify the search by forming clusters that contain three subsequent peaks. The magnitudes of these peaks are evaluated; if the middle peak (MHP) is higher than the two accompanying peaks (to its left and right), then the GMPP is considered to be found, and the search is halted. If the middle peak is lower than the two accompanying peaks, then the algorithm assumes that GMPP does not exist within that particular cluster. Then, a successive cluster is formed, and this new cluster is scrutinised in the same manner as the previous one. Although this procedure significantly improves the tracking speed, it will fail for the $P-V$ curve that exhibits several clusters and the GMPP is not present in the first cluster. This finding was proven by Kermadi *et al.* [25], and an example is shown in Fig. 1. As can be observed, cluster 1 has a lower MHP than cluster 2. However, the algorithm by Tey and Mekhilef assumes MHP2 is the GMPP because it considers the condition for the MHP within the cluster is met. Thus, the search is halted and

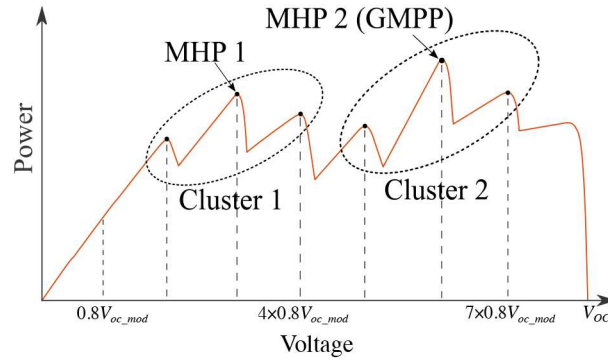


Fig. 1 P - V curve with two clusters and two MHPs [25]

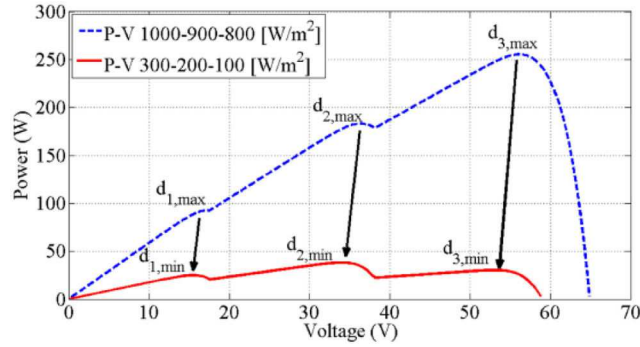


Fig. 2 Duty cycle ranges of the local power peaks [28]

the scan will not go beyond MHP1. The result is obviously wrong because the true GMPP is located in cluster 2, i.e. MHP2.

In [26], Kouchaki *et al.* improved the algorithm proposed in [1] by performing two-step modifications: first, instead of setting $\phi \leq 0.8V_{oc}$, it checks all the multiples of $0.8 \times V_{oc}$ to locate all the possible local MPPs. Then, the precise condition is applied to detect the occurrence of partial shading by using the following equation:

$$\frac{\Delta I}{I_c}(\text{Uniform}) \geq \frac{\Delta I}{I_c}(W = 1000 \text{ W/m}^2)_{ref} \quad (1)$$

$$> \frac{\Delta I}{I_c}(\text{partially shaded})$$

Despite the improvement, this method exhibits several drawbacks. If the local peaks start to drift away from the $0.8 \times V_{oc}$ positions, then the algorithm may fail to converge to GMPP. Also, the algorithm has to continuously update V_{oc} , which is troublesome if the number of series modules is high.

In [27], Murtaza *et al.* proposed a more practical GMPPT algorithm, where they consider multiple bypass diodes (typically three, in a real product) to be present inside a single module. This situation is in contrast to [1], which assumed only one bypass diode per module. Thus, the step size used for scanning is modified as follows:

$$\Delta V = \frac{V_{oc_array}}{N_s} \times \frac{1}{N_{BD,M}} \quad (2)$$

where N_s is the number of series-connected modules and $N_{BD,M}$ is the number of bypass diodes in the module. While the peak detection process is similar to [1], the search zone is set from zero to V_{Lim} , where

$$V_{Lim} = ((N_s N_{BD,M}) - 1) \Delta V + \Delta V_{1st} \quad (3)$$

V_{Lim} is near the open-circuit voltage of the array, i.e. V_{oc_array} . The search starts from $\Delta V_{1st} = \Delta V/2$, then the voltage is increased by ΔV in every step as follows:

$$\frac{\Delta V}{2}, \frac{3\Delta V}{2}, \frac{5\Delta V}{2} \dots \frac{n\Delta V}{2} \quad [n \text{ is an odd number}] \quad (4)$$

Similar to [27], the work in [28] also considered the presence of several bypass diodes within the module level. However, instead of controlling the voltage, it regulates the duty cycle to scan the multiples of $0.8 \times V_{oc}$ vicinities. Furthermore, the authors considered that solar irradiance varies from 100 to 1000 W/m² throughout the day. The authors conducted experiments with three modules; thus, the search ranges were categorised into three sections, namely, $[d_{1,min} \text{ to } d_{1,max}]$, $[d_{2,min} \text{ to } d_{2,max}]$, and $[d_{3,min} \text{ to } d_{3,max}]$, as shown in Fig. 2.

The following expressions are used to determine the search range for the duty cycle for every peak:

$$d_{1,2,3}(\min) = \frac{1}{1 + \sqrt{(V_{MPP}/nI_{MPP})/R_L}} \Big|_{n=3,2,1} \quad (5)$$

$$d_{1,2,3}(\max) = \frac{1}{1 + \sqrt{(qV_{MPP}/nI_{MPP})/R_L}} \Big|_{n=3,2,1}$$

where q is the ratio between min and max irradiance values and R_L is the resistive load. In this approach, the search range for the duty cycle is significantly narrowed by considering the peaks located at $0.8 \times V_{oc}$ multiples. This algorithm regulates the duty cycle, which is why the search within the selected range is quite fast.

In a separate work, Ahmed and Salam [29] concluded that the $0.8 \times V_{oc}$ method is applicable for mono- and polycrystalline only; it does not include thin film and other types of modules. They highlighted several points that led to further improvements. First, the assumption is that the peaks are in the vicinity of multiples of $0.8 \times V_{oc_mod}$. While this idea is valid for a small number of local peaks (3 or 4 in total), it does not hold true otherwise. The authors proved that as the number of peaks grows, the positions of the peaks drift away from the multiple of $0.8 \times V_{oc}$. Thus, under such circumstances, the algorithm cannot guarantee that the GMPP is tracked correctly. On the basis of this finding, a compensation variable (α) is incorporated into the formula; α varies from 0.8 to 0.97, depending on the shading intensity.

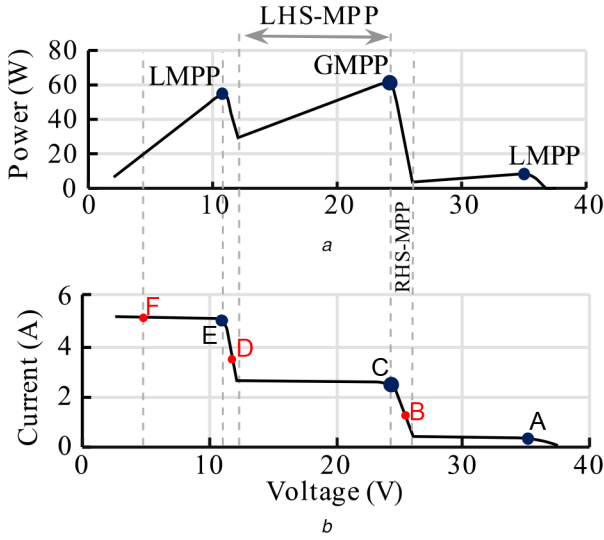


Fig. 3 Tracking mechanism of ST GMPPT [31]
(a) Power-Voltage curve, (b) Current-Voltage curve

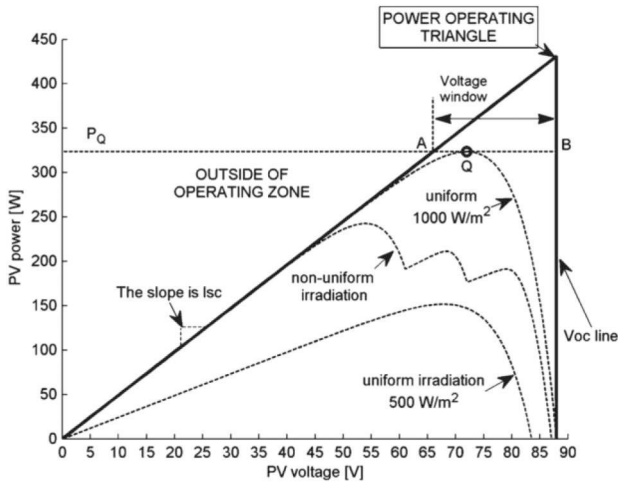


Fig. 4 POT for VWS [6]

2.1.2 Direct duty cycle method: In [30], Kazmi *et al.* proposed a variation of P&O for partial shading by using the direct duty cycle method. It measures V_{oc} and I_{sc} by setting the duty cycle to 0 and 1, respectively. To calculate the voltage and current at MPP (V_{MPP} and I_{MPP}), two multiplying factors are introduced, i.e. M_v and M_i . Initially, the values for M_v and M_i are set to 0.8 and 0.9, respectively. These values are returned periodically once the computation for the MPPT is completed. The crucial part of this algorithm is the one-step control, in which the duty cycle at MPP (D_m) is determined as follows:

$$D_m = 1/(1 + \sqrt{R_m + R_{load}}) \quad (6)$$

Under normal conditions, the MPP is detected directly by the one-step control. The MPP and duty cycle are recorded, and the values of M_v and M_i are updated. Under partial shading, the algorithm scans the P - V curve in the range of duty cycle 0 to 1. It starts from the previously saved duty cycle value D_m , i.e.

$$d(n) = D_m + \Delta d \quad (7)$$

$$\Delta d = (-1)^{itr} \times itr \times d_0$$

where itr is the iteration number (1, 2, ..., n), and d_0 is the duty cycle perturbation. After the scan is completed, the sample points are compared, and the one that gives the highest power value is chosen. From that sample, the P&O is initiated to maintain the operating point at GMPP.

2.1.3 Shade-tolerant method: Hosseini *et al.* [31] proposed the shade-tolerant (ST) GMPPT by incorporating additional codes into the P&O. The authors observed that the voltage interval in the P - V curve can be categorised into two regions: the left-hand side of MPP (LHS-MPP) and the right-hand side of MPP (RHS-MPP), as shown in Fig. 3a. The LHS region occupies the majority of the interval. Thus, the objective of the ST method is to avoid scanning the LHS-MPP region.

When partial shading occurs, the algorithm starts by looking for the rightmost peak using P&O. After the MPP (point A) is reached, as shown in Fig. 3b, the reference current is perturbed to skip the LHS-MPP region and operates at the RHS-MPP region of the subsequent higher current step (point B). Starting from that point, the second MPP is tracked using P&O (point C). The perturbation in the reference current is realised using the following equation:

$$I_{ref}(k) = (1 + \alpha)I_{ref}(k - 1) \quad (8)$$

where α is fixed at 0.3 to ensure that the operation is within the RHS-MPP region when shifting to the next LMPP. Fixing α at 0.3 was based on the assumption that the local power peaks always occur at integral multiples of $0.8 \times V_{oc}$ (observation 1 about the $0.8 \times V_{oc}$ method). Doing so ensures that the LMPP tracking will be faster and P_{GMPP} and V_{GMPP} are checked and updated if necessary after each power peak tracking. The same procedure is followed for the subsequent current steps (points D and E) until the highest one is reached, i.e. the leftmost side of the curve (point F). In the end, generic P&O is initialised from the recorded GMPP.

2.2 GMPPT with skipping mechanism

The GMPPT using the skipping mechanism is the recent innovation to improve the conventional MPPT under partial shading conditions. It involves a skip procedure to identify areas that do not contain the GMPP and to avoid scanning them. These unnecessary areas are not searched; thus, the search space is reduced, thereby improving the tracking speed significantly [10].

2.2.1 Voltage window search (VWS): VWS uses the concept of a power operating triangle (POT) to identify a voltage window (VW) range $[V_{min}, V_{max}]$. Within this window, the GMPP is guaranteed to be found [6]. The upper limit of the range (V_{max}) is fixed at the maximum possible GMPP voltage. The key idea is to narrow the lower limit (V_{min}) so that the search is minimised. Extensive simulation by Boztepe *et al.* [6] concluded the maximum GMPP voltage never exceeds $0.887 V_{OC}$, where V_{OC} is the array open-circuit voltage. Thus, V_{max} is fixed at $0.9 V_{OC}$. A fixed voltage step (ΔV_{GSTEP}) is used to scan the VW. After an extensive statistical study, Boztepe *et al.* [6] found that the optimum value of ΔV_{GSTEP} should be approximately the voltage difference between two adjacent local power peaks, i.e. within the range $[0.5 V_{ocp}, V_{ocp}]$, where V_{ocp} is the open-circuit voltage of a single part of a PV module under the standard test conditions (STCs). This finding is used to improve the accuracy of observation 2 about the $0.8 \times V_{oc}$ method.

Fig. 4 depicts the POT. It is delimited by the x-axis, the straight lines of V_{OC} and the power line, whose slope is the short-circuit current (I_{sc}) at STC. This hypotenuse slope is progressively reduced using the current measurements during the search for GMPP. The POT is employed to update V_{min} and reduce the VW. For an initial operating point Q, its power P_Q is stored in P_{store} , V_{max} remains constant at $0.9 V_{OC}$, while V_{min} is calculated as follows:

$$V_{min} = \frac{P_{store}}{I_{POT}} \quad (9)$$

where P_{store} is the memory space reserved to store the best power recorded and I_{POT} is the slope of the POT hypotenuse, which equals to I_{sc} in the POT. Point A is the intersection point after

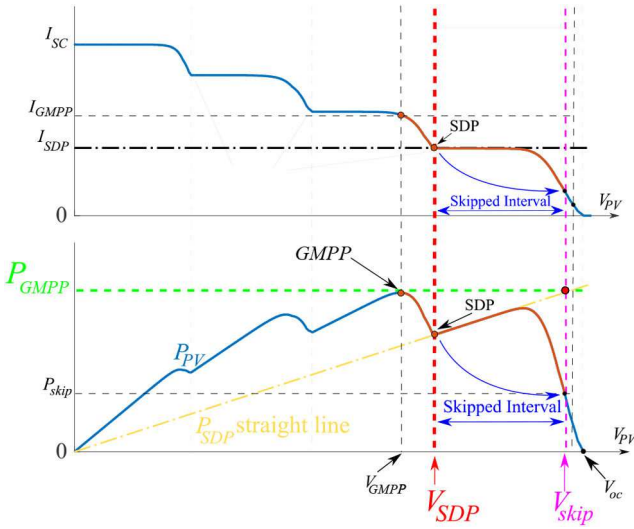


Fig. 5 Description of the skipping mechanism by SSJ

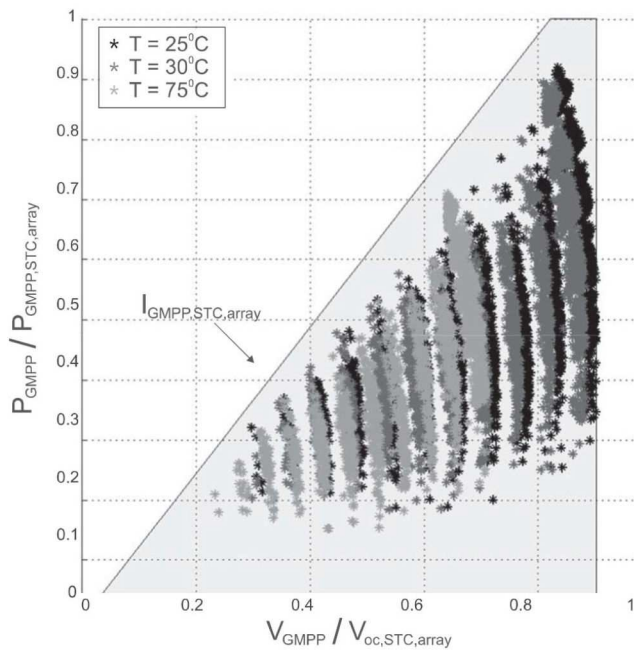


Fig. 6 GMPP distribution in the normalised plane with different values of temperature and random irradiances [8]

projecting point Q horizontally into the POT hypotenuse. Hence, $V_{\min}$ is obtained as the voltage of point A. Within the voltage interval $[0, V_{\min}]$, the power line of the POT hypotenuse is always higher than the PV power. The power for any operating point within $[0, V_{\min}]$ can be guaranteed to never exceed P_Q ; hence, the voltage interval $[0, V_{\min}]$ can be exempted from being searched. Thus, $V_{\min}$ is set as the lower limit of the VW.

2.2.2 Search-skip-judge (SSJ): The P - V curve shown in Fig. 5 is considered to understand the mechanism involved in SSJ [7]. The objective of SSJ is to minimise the search region by identifying the sections in which the search can be omitted. First, from the lower end of the curve, the nearest LMPP is determined by InCond. Then, its corresponding section dividing point (SDP) is calculated. The SDP is a local minimum point (V_{SDP}, P_{SDP}), located at the right side of the power peak; it is defined as the voltage in the I - V curve at which the irradiance (thus the current) steps due to partial shading occurrence. For instance, in Fig. 5, the I - V curve is composed of four steps due to the shading pattern of four irradiance levels; the point that divides the step is the SDP. The following procedure determines the skipped interval; after initialisation, the operating point climbs to the nearest power peak

by using the InCond and records the voltage and current measurements (GMPP point in Fig. 5). Then, the operating point is perturbed forward towards the right side until the SDP is tracked. When the voltage of the operating point (V_{PV}) is higher than V_{SDP} , the current I_{PV} is always lower than I_{SDP} . Therefore, within the voltage interval $[V_{SDP}, V_{skip}]$, the operating power P_{PV} is always below the P_{SDP} straight line. V_{skip} is the voltage at which the P_{SDP} straight line intersects with the value of P_{GMPP} . $P_{GMPP} = P_{SDP}$ at V_{skip} ; therefore, it follows that the operating power would never be higher than P_{GMPP} within the interval $[V_{SDP}, V_{skip}]$. Clearly, in this condition, P_{GMPP} is guaranteed to be highest within that interval. Hence, the upper bound of the skipped voltage interval V_{skip} can be deduced as follows:

$$V_{skip} = \frac{P_{GMPP}}{I_{SDP}} \quad (10)$$

It is guaranteed that no other peak is higher than P_{GMPP} ; thus, the interval $[V_{SDP}, V_{skip}]$ can be exempted from being explored. SSJ is particularly fast when the GMPP is located at the lower end of the P - V curve; however, at the upper end, it is much slower because the search and skip process needs to be repeated for several lower voltage peaks before reaching the GMPP.

2.2.3 Maximum power trapezium (MPT): The MPT relies on the array open-circuit voltage at STC ($V_{OC,STC}$) and the array short-circuit current at STC ($I_{MPP,STC}$) data [8]. To avoid missing the GMPP, the P - V curve is scanned using a dynamic time step (T_{max}), which is carefully selected to ensure that the maximum voltage step equals the minimum difference between two adjacent local MPPs ($V_{\min,MPPT}$). The value of $V_{\min,MPPT}$ can be chosen as the V_{MPP} at 100 W/m^2 and 75°C [8]. Therefore, T_{max} can be determined using the following formula:

$$T_{max} = \frac{V_{\min,MPPT} \times C_{PV}}{I_{C,max}} \quad (11)$$

where C_{PV} is the DC bus capacitor of the grid-tied inverter and $I_{C,max}$ is the maximum DC bus current. Therefore, if the DC bus current I_C is at its maximum value, then the corresponding voltage step of the MPT algorithm (V_{GSTEP_MPT}) after each period T_{max} is equal to $V_{\min,MPPT}$. However, in case of I_C is not maximum, the V_{GSTEP_MPT} can be calculated as follows:

$$V_{GSTEP_MPT} = \frac{I_{C,max} \times T_{max}}{C_{PV}} \quad (12)$$

In this case, the voltage difference (V_{GSTEP_MPT}) between any two successive samples (after T_{max}) is lower than the optimum value $V_{\min,MPPT}$. Consequently, this condition may result in sluggishness in curve scanning. The same authors conducted a general statistical study and found that for random partial shading patterns, the GMPP distribution is enclosed within a trapezoidal area in the P - V plane, as shown in Fig. 6. All GMPPs are limited by a vertical segment at $0.9 V_{OC,STC}$. To the left of this vertical line, the GMPPs are confined within the slope characterised by the GMPP current at STC ($I_{GMPP,STC}$) and limited by the GMPP power at STC ($P_{GMPP,STC}$). This closed area is called the MPT. The GMPTT is guaranteed to be within the MPT, which is why scanning points outside it is not needed.

Fig. 7 depicts the procedure to determine $V_{\min}$, which is the lower bound of the voltage range to be searched. Projecting Point A (extending it to the left of the curve) into P_{MPT} obtains the value of the voltage $V_{\min}$ at which the projected power value at point A (P_A) intersects with P_{MPT} (i.e. $P_A = P_{MPT}$). Recall that P_{MPT} is the hypotenuse line of the MPT. P_{MPT} is always higher than P_{PV} within the voltage interval $[0, V_{\min}]$. The power value is guaranteed

to never exceed P_A within $[0, V_{\min}]$; therefore, this voltage interval can be exempted from being searched. Hence, $V_{\min}$ can be calculated using the following expression:

$$V_{\min} = \frac{P_{\text{GMPP}}}{I_{\text{MPP,STC}}} \quad (13)$$

Despite the apparent similarities between the MPT [8] and the VWS [6], they differ in two aspects. The first is that the hypotenuse slope ($I_{\text{MPP,STC}}$) of the MPT (with vertices a, b, c) is smaller than the hypotenuse slope ($I_{\text{SC,STC}}$) of the initial POT (with vertices a, b, d), as depicted in Fig. 8. For a similar case, $V_{\min, \text{MPP}}$ in [8] is higher than $V_{\min, \text{SC}}$ in [6] because I_{SC} is always higher than I_{MPP} regardless of the irradiance. Thus, the exempted interval for MPT $[0, V_{\min, \text{MPP}}]$ is larger than that using POT $[0, V_{\min, \text{SC}}]$. In other words, the interval to be scanned with MPT ($[V_{\min, \text{MPP}}, V_{\max}]$) is smaller than POT ($[V_{\min, \text{SC}}, V_{\max}]$).

The second difference is that the MPT triangle is fixed with time during the search for GMPP. The POT is dynamic. Its hypotenuse slope gradually decreases as the search progresses. Although the skipped interval of the initial POT is smaller than MPT, overall, the dynamism of the former results in a higher skipped interval. This latter is one of the factors that explain the improved convergence speed of VWS over MPT.

2.3 Soft computing GMPPT

SC GMPPT capitalises on the vast power of modern computers to search for a global peak. It mathematically models the behaviour/phenomena observed in nature by imposing systematic movement of searching agents in the search space. The standard procedures of SC are summarised as follows: (i) an objective function is defined to quantify the fitness of each candidate solution, (ii) a number of candidate solutions (agents) with predefined initial positions are defined, (iii) the fitness of each agent according to the objective function is evaluated, and (iv) the position of each agent is updated using an optimiser function. Despite their advantages, particularly in handling partial shading, all SC techniques exhibit one common drawback: sluggishness.

Most of the SC GMPPTs were extensively reviewed (Table 1 in the Introduction); thus, only a brief description is given here. The SC techniques are classified into three main categories, as shown by the family-tree diagram in Fig. 9. The first is the family of brain-inspired computing methods, which comprises artificial neural network (ANN) [32], fuzzy logic control (FLC) [33], machine learning [34], and deep learning [35]. The second is chaos optimisation [36] and the third one is the family of metaheuristic algorithms. The metaheuristic methods can be divided further based on the source of inspiration for the algorithm.

2.3.1 Swarm intelligence: Swarm intelligence imitates the behaviour of creatures such as animals, insects, or fishes in search for food. The equations used for the agents' movement during the

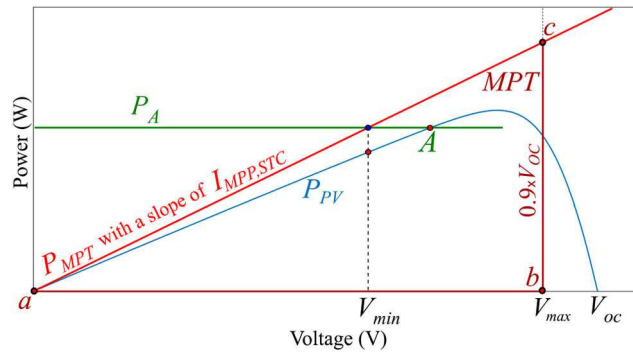


Fig. 7 Zones and the maximum voltage and current

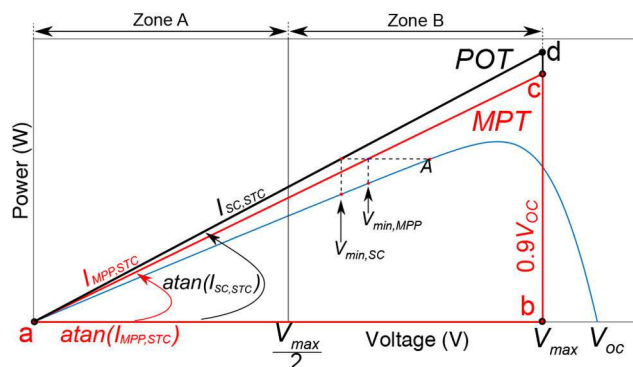


Fig. 8 Difference between POT and MPT in P-V graphic

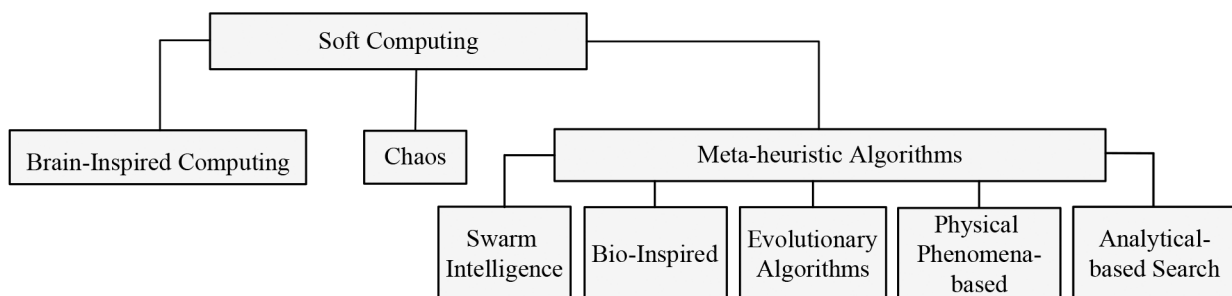


Fig. 9 Family tree for GMPPT techniques based on SC approach

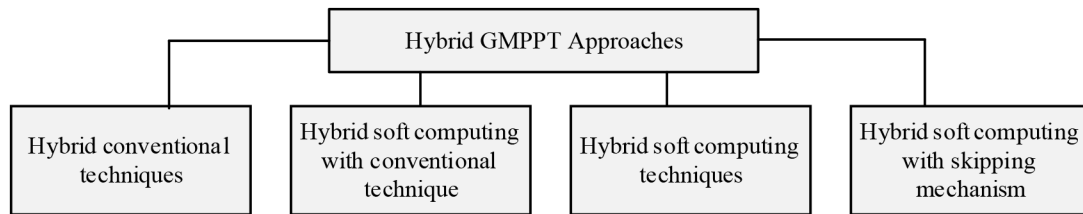


Fig. 10 Taxonomy of hybrid MPPT approaches for PV systems

optimisation process are inspired by the interactive behaviour of the swarm while looking for food. Among the popular algorithms adopted for GMPPT are PSO [37, 38], ACO [39], ABC [40], FSO [41], glowworm swarm optimisation [42], human psychology optimisation, fruit fly optimisation algorithm [43], BSA [44], FFA [45, 46], GWO [47], krill herd algorithm [48], whale optimisation algorithm (WOA) [49], and dragonfly optimisation algorithm [50].

2.3.2 Bio-inspired algorithms: These algorithms are also based on nature. However, they do not exhibit the swarm behaviour to find the optimal solution for an optimisation problem. For example, the flower pollination algorithm (FPA) [51] is inspired by the mechanism of the pollination of flowering plants. FPA considers the cross-pollination or allogamy process as a global search and the pollen-carrying pollinators performing the Levy flights. Self-pollination is considered as local pollination or as a local search. Thus, no interaction occurs between agents, such as in the swarm intelligence (SI) techniques. CS [52] mimics the breeding behaviour of cuckoo birds. The sample points are the cuckoo's eggs laid in the nest of a different species of bird. On the basis of the outcomes of the samples, good eggs and rotten eggs are identified. Afterwards, rotten eggs are replaced by better eggs, which mimic the quality of other good eggs. Mathematically, the Levy distribution is used to reproduce these new eggs. Over the iterations, all the eggs attain better properties or move towards the optimised solution. Other examples of bio-inspired MPPT algorithms include shuffled frog leap algorithm (SFLA) [53] and crowded plant height optimisation [54].

2.3.3 Evolutionary: Evolutionary algorithms such as GA [55, 56], DE [57, 58], evolutionary programming [59], and immune algorithm [60] are widely used for MPPT. They utilise key genetic operators such as crossover and mutation. Certain techniques, such as GA, rely primarily on a crossover, while others, such as DE, depend heavily on mutation. For MPPT applications, the EA algorithms usually require a longer time to converge to GMPP.

2.3.4 Physical phenomena-based: Certain phenomena in physics inspire researchers to derive MPPT algorithms. One such phenomenon is simulated annealing (SA) [61]. It mimics the annealing process when a metal is heated up to the molten state and then cooled down to become stable. This phenomenon is adapted in the MPPT problem by representing the PV voltage and power as the metal's temperature and energy, respectively. The main parameters to be observed are initial temperature, final temperature, and cooling rate. Several perturbations on the operating temperature are used to check the corresponding energy of the metal on the basis of the essential probability criterion. Convergence is achieved when the temperature variation is at the highest, while the change in energy is at its lowest. Other physical-based algorithms used for MPPT are radial movement optimisation [62], memetic slap swarm algorithm [63], black hole algorithm [64], central force algorithm, and firework algorithm (FWA) [65].

2.3.5 Analytical-based GMPPT: Several researchers proposed GMPPT based on analytical formulas. This method provides closed-form solutions that immediately identify the MPP [66]. Examples of analytical GMPPT are the Jaya algorithm (JA) [67], binary search [68], generalised pattern search optimisation algorithm [69], Lipschitz optimisation [70], mine blast algorithm, and the teaching-learning based optimisation technique [71]. In general, their optimiser functions are purely mathematical formulas

and not inspired by any particular biological or physical evolution. The main shortcomings of these techniques are heavy mathematical computation and the need for prior knowledge on the module parameters. In addition, it requires frequent interruptions to measure the voltage and current of the system.

Despite the widespread utilisation of SC MPPT techniques, several issues need to be considered. The first is the inability to converge to GMPP. Failure to reach GMPP results in high power loss. The second is the comparatively long convergence time. An extended search duration increases the instability of the system and reduces the overall efficiency. Unfortunately, these issues frequently arise due to the following reasons: (i) inappropriate selection of the optimisation parameters, (ii) an unsuitable number of agents, and (iii) wrong initialisation of the searching agents. A systematic method to determine the optimal values of these parameters is lacking; thus, these optimal values are mainly obtained through experience and an extensive trial-and-error process.

2.4 Hybrid GMPPT

Generally, tracking accuracy comes at the expense of sluggish tracking speed. As a compromise, several researchers combined two or more techniques to extract the desirable features of one technique while compensating for the weakness of the other. These are known as hybrid MPPTs; examples are shown by the family-tree depicted in Fig. 10.

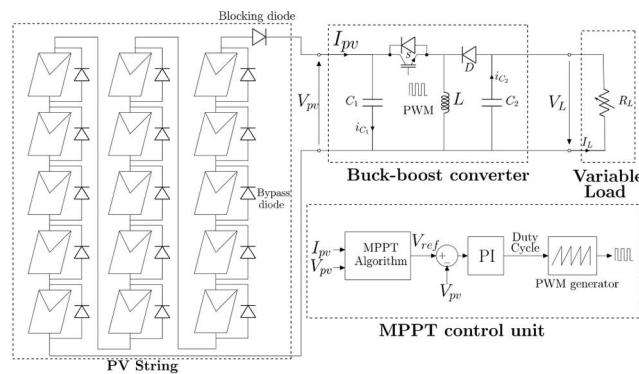
The different types of hybridisation published in the literature are reviewed here.

2.4.1 Hybrid between two conventional MPPTs: This approach combines two conventional MPPTs to improve tracking performance. Sher *et al.* [72] proposed a hybridisation between fractional short circuit current (FSCC) and P&O (FSCC-P&O). It exploits the fast-tracking of FSCC during the initial stage and the accurate tracking of P&O once the MPP is reached. Hou *et al.* [42] utilised a similar concept, but instead of FSCC, fractional open-circuit voltage (FOCV) is used. In [43], FOVC is combined with a current sensorless method with automodulation, while in [44], InCond (FOCV-InCond) is used. Generally, the hybrid of two conventional MPPT results has faster convergence to GMPP, but the reliability is very low, particularly in tracking multiple peak curves. Hence, they are not suitable for handling partial shading conditions.

2.4.2 Hybrid between SC and conventional MPPT: The SC is known for its accuracy, while the conventional MPPT has a fast response. For this reason, the former is used to locate the vicinity of the GMPP in the first stage of the tracking. Afterwards, the conventional MPPT will ensure that small variations in the GMPP position is continuously tracked. This dual-stage for tracking is the best option for PV systems that are subjected to variable insolation conditions (uniform insolation and partial shading). Numerous schemes belong to this category: FLC with P&O (FLC-P&O) [73]; FL-INC [74, 75]; GA-P&O [76], ANN-P&O [77], FFA-INC [78], modified BA-P&O [79], hybrid PSO-P&O in all of [80–82], PSO-InCond [83], GWO-P&O [84], FWA-P&O [85], FPA-P&O [86], and ACO-P&O [87]. Overall, the tracking performance is increased in terms of accuracy, but the sluggishness remains because of the metaheuristic element embedded in these algorithms.

Table 2 Selected GMPPT algorithms for the critical evaluation test

GMPPT technique	Abbreviation	Category	Ref	Remarks
modified incremental conductance	MIC	$0.8 \times V_{oc}$ method	[2]	a fast GMPP tracking method that employs clusters of three consecutive peaks
shade tolerant	ST	$0.8 \times V_{oc}$ method	[31]	a mathematical expression derived from the $0.8V_{oc}$ model is used for the fast scanning of the local peak areas using P&O
voltage window search	VWS	skipping mechanism	[6]	a POT is employed to skip a certain area in the search range from being explored
search skip judge	SSJ	skipping mechanism	[7]	an InCond is employed in conjunction with a skipping method using LMPPs and SDPs to avoid scanning the area free from the GMPP
maximum power trapezium	MPT	skipping mechanism	[8]	A MPT is employed to skip the search on certain (GMPP-free) intervals
flower pollination algorithm	FPA	SC	[51]	A modified bio-inspired search ensures fast-tracking and enhanced accuracy
modified CS	MCS	SC	[100]	a modified bio-inspired search improves the tracking speed and accuracy
effective hybrid PSO–SSJ	EHY	hybrid	[25]	PSO works in conjunction with the SSJ mechanism to enhance the tracking performance and accuracy

**Fig. 11** Simulation setup used for evaluation

2.4.3 Hybrid between two SC MPPTs: Usually, SC techniques are accurate, but they suffer from sluggishness. In contrast, certain algorithms, namely, FL and ANN [88], are high-speed. Therefore, several researchers proposed combinations between two SC methods to increase the speed; examples are PSO–FLC [89] and GWO–FLC [90]. Metaheuristic search is used in the first stage to identify the vicinity of the global peak; after that, the FL is used to track the GMPP accurately. An improved PSO with brain emotional learning-based intelligence (IPSO–BELBI) is used for the same reason [91]. Another group of researchers opted to combine two or more metaheuristic search methods: WO–DE [92], PSO–SFLA [93], WOA–SA [94], FLC–FFA [95], and DE–PSO [15]. Others use PSO with FSA using an extended memory strategy [96], Gaussian process regression–JA with PSO [97], and ANN–FLC [77].

2.4.4 Hybrid between SC and skipping mechanism: In [25], an effective hybrid algorithm (EHY) that combines a skipping mechanism (SSJ) with a SC algorithm (PSO) was proposed to improve the convergence speed and tracking accuracy. The PSO particles utilised SSJ to reduce the search space by skipping the area that does not contain the GMPP. Hybridisation reduces the convergence time and improves overall tracking performance considerably.

3 Benchmarking of selected GMPPTs

As mentioned in the Introduction, the performance of various GMPPTs under partial shading conditions needs to be evaluated and benchmarked. Thus far, no such work has been carried out; this effort is the contribution to fill this knowledge gap.

3.1 PV system setup

The critical evaluation of MPPT from each category is carried out in the MATLAB–Simulink environment. The competing GMPPT

techniques are given in Table 2. The values of the control or optimising parameters for each technique are directly taken from the respective references. For a fair assessment, they are tested in the same setup and subjected to the same test conditions. The same control strategy, i.e. indirect control, is implemented [98]. In this scheme, the MPPT generates a reference voltage, and the duty cycle of the converter is adjusted such that the PV output voltage matches this reference voltage [99]. The complete block diagram of the test setup with its associated control block is shown in Fig. 11. The buck–boost power converter is selected due to its ability to operate in the full voltage range regardless of the load value connected to its output. The following converter parameters are selected to ensure the operation in continuous conduction mode: $L = 1$ mH, $C_1 = 470$ μ F, $C_2 = 220$ μ F, with a 50 kHz switching frequency.

3.2 Partial shading test profile

A test sequence that consists of 16 complex partial shading patterns is successively imposed on the algorithm. These patterns are fed to a string of 15 series-connected PV modules, as shown in Fig. 12. The shading arrangement and intensity of the shaded modules are deliberately designed to ensure the GMPPs appear within four possible regions on the P – V curve: at the leftmost (zone 1), middle left (zone 2), middle right (zone 3), and rightmost side (zone 4). These regions are shown in Fig. 13. During the 16-s test sequence, the shifting from the shading pattern to another is performed at a 1-s time interval. The purpose of the shift is to emulate the real workings of the system, where the irradiance and shading dynamically change with time. The transitions are carefully selected such that for every shift, the operating point moves from one zone to another. The tool proposed in [101] is used to generate the partial shading patterns. All transitions of the GMPP that existed in the test profile are represented in the P – V plane, as shown in Fig. 14. All the shifts labelled in Fig. 14 are detailed in Table 3.

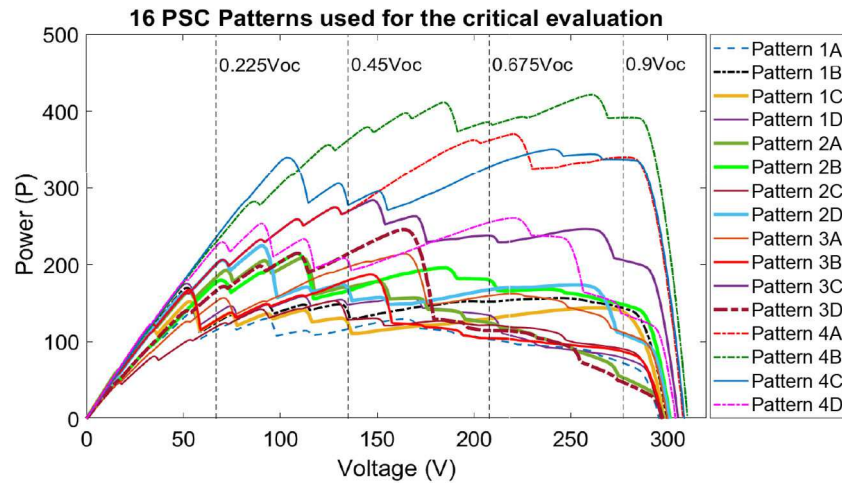


Fig. 12 Sixteen complex shading patterns composing the test profile

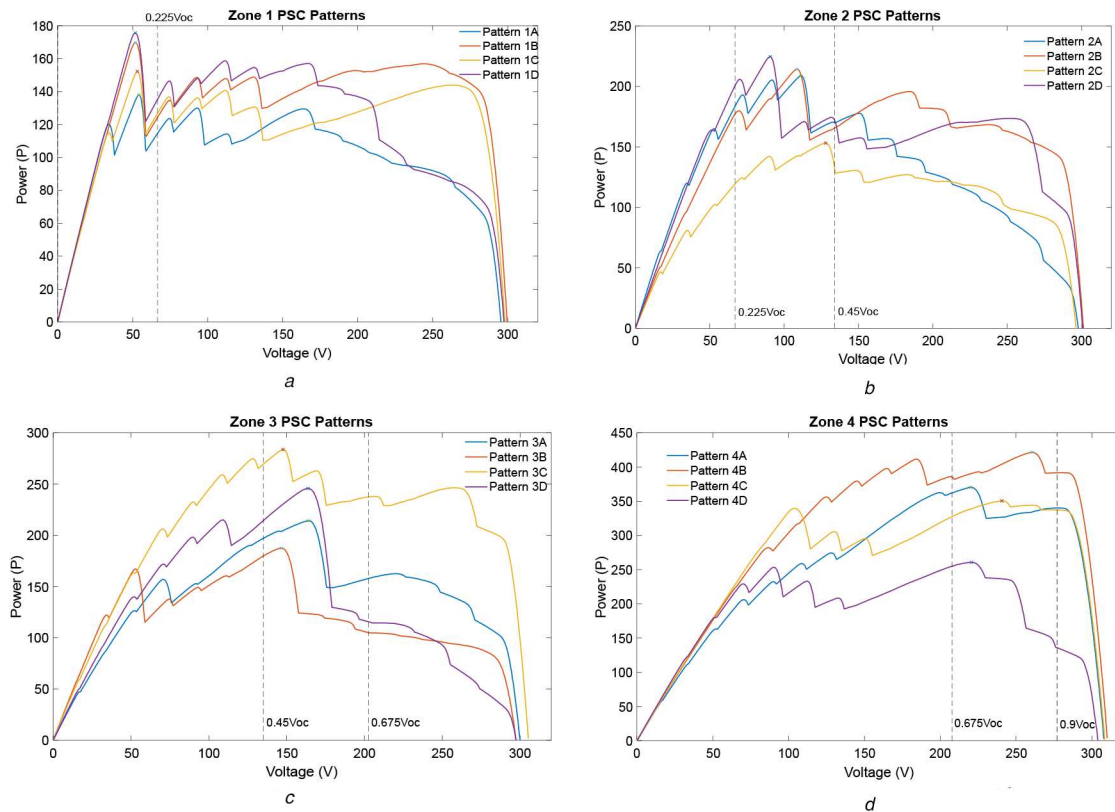


Fig. 13 P - V curve is subdivided into four zones

(a) GMPP appears in zone 1 $[0, 0.225V_{oc}]$, (b) GMPP appears in zone 2 $[0.225V_{oc}, 0.45V_{oc}]$, (c) GMPP appears in zone 3 $[0.45V_{oc}, 0.675V_{oc}]$, (d) GMPP appears in zone 4 $[0.675V_{oc}, 0.9V_{oc}]$

3.3 Results and discussion

The evaluation is based on four main criteria. The first is the ability of the GMPPT to converge to the correct GMPP. This criterion is the most crucial because it determines the reliability and robustness of the algorithm. The second is the convergence time, i.e. the total amount of time that is needed to reach the GMPP. The third is the voltage track, which is the path length that the operating point has to travel along the voltage axis until it converges to GMPP. The fourth is the transient efficiency (η), which is defined as the MPPT efficiency during the transition from one shading pattern to another [10]. In this section, only results that relate to the critical findings are included. Other results, including the figures of power and voltage waveforms (Figs. 15–22) and numerical values of evaluation criteria for all GMPPT methods (Table 4).

3.3.1 Success rate: Fig. 23 summarises the rate of convergence for the 16 shading patterns. The skipping-based algorithms,

namely, VWS, SSJ, and ST, have perfect success rates. For the power waveform profile for VWS shown in Fig. 24a, the operating point converges to GMPP for every shading pattern. The shading intensities and transitions are carefully designed to include the GMPP in all possible zones of the P - V curve, thereby suggesting that VWS, SSJ, and ST are very robust and reliable. In contrast, the modified InCond (MIC) exhibits a low success rate (62%). The failure of the algorithm is due to the premature search termination for multiple clusters, as described in Section 2. For certain shading patterns (4D, 4B, 4C, 2C, 3A, and 3B), the GMPP lies beyond the first cluster; as a result, the MIC fails to detect it [25]. This condition can be clearly observed in the power (versus time) profile shown in Fig. 24b. Furthermore, the MPT, which also utilises the skipping mechanism, does not achieve the expected 100% convergence. It misses the GMPP for patterns 1C and 1B due to the large voltage step used for the scan, as identified by Kermadi *et al.* [10]. Nonetheless, the tracking success of MPT can be improved if a simple fine-tuning of the step is considered [6].

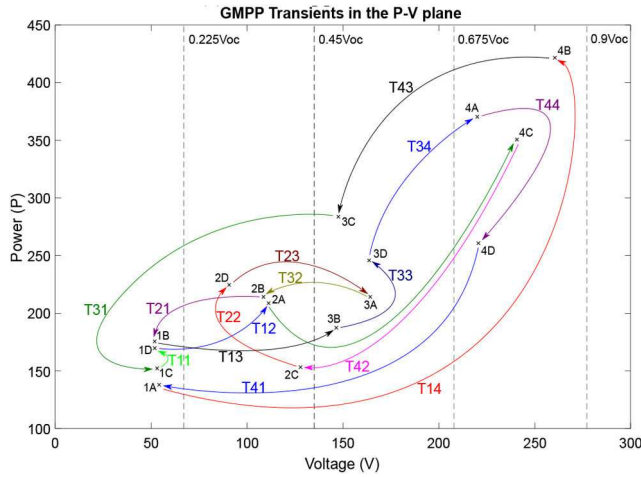


Fig. 14 Plot of all GMPP transitions in the P - V plane

Table 3 GMPP shifts comprised in the 16 s test profile

Label	Time, s	Initial GMPP zone	Final GMPP zone	Pattern transition
T41	1	4	1	4D to 1A
T14	2	1	4	1A to 4B
T43	3	4	3	4B to 3C
T31	4	3	1	3C to 1C
T11	5	GMPP remains in zone 1		1C to 1D
T12	6	1	2	1D to 2A
T24	7	2	4	2A to 4C
T42	8	4	2	4C to 2C
T22	9	GMPP remains in zone 2		2C to 2D
T23	10	2	3	2D to 3A
T32	11	3	2	3A to 2B
T21	12	2	1	2B to 1B
T13	13	1	3	1B to 3B
T33	14	GMPP remains in zone 3		3B to 3D
T34	15	3	4	3D to 4A
T44	16	GMPP remains in zone 4		4A to 4D

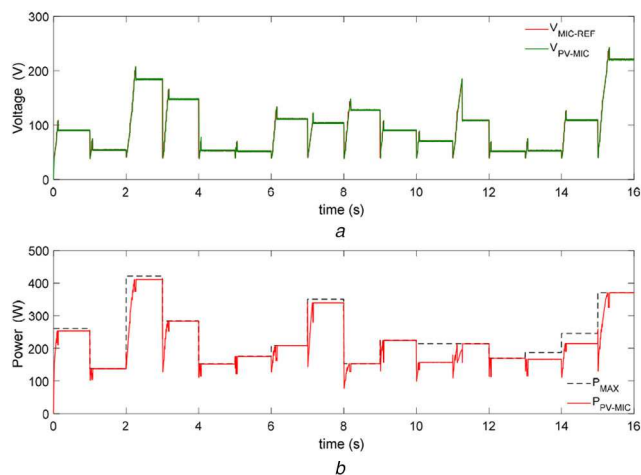


Fig. 15 Power and voltage fluctuations of the MIC algorithm [2]
(a) Voltage fluctuations, (b) Power fluctuations

However, because the rules for the evaluation are based on codes published in the original paper [8], the improvement suggested by Boztepe *et al.* [6] cannot be taken into account.

The SC-based methods (represented by MCS and FPA) appear to have very low success rates (56 and 50%, respectively). This result is in contrast to the expectation that the metaheuristic search is able to locate the GMPP regardless of the shape of the P - V curve [13]. One probable explanation for this discrepancy can be

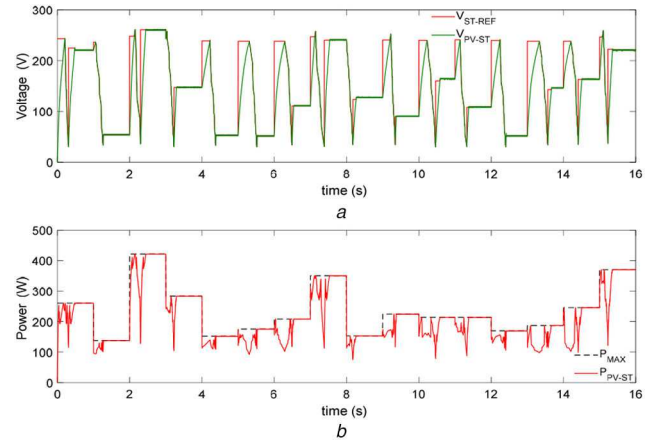


Fig. 16 Power and voltage fluctuations of the ST algorithm [31]
(a) Voltage fluctuations, (b) Power fluctuations

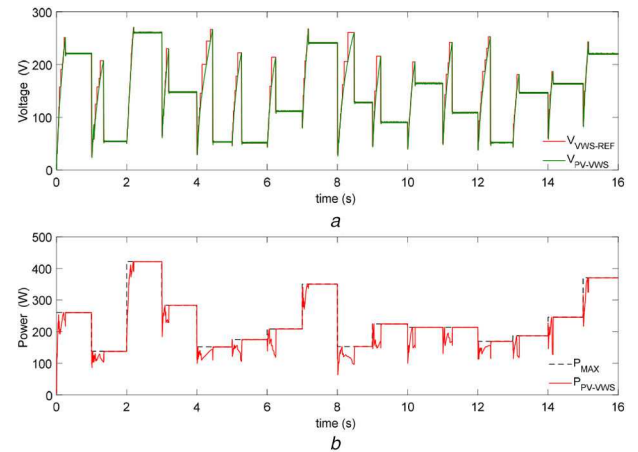


Fig. 17 Power and voltage fluctuations of the VWS algorithm [6]
(a) Voltage fluctuations, (b) Power fluctuations

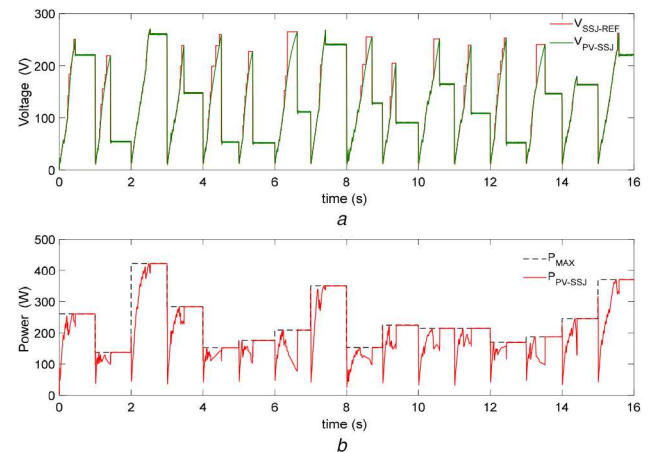


Fig. 18 Power and voltage fluctuations of the SSJ algorithm [7]
(a) Voltage fluctuations, (b) Power fluctuations

attributed to the selection of the optimiser parameters. The selected parameter values for the simulation are likely to be high; if that was indeed the case, then the search agents are forced to jump towards the best positions with a large voltage step, thereby missing the GMPP. Finally, the EHY (which represents the hybrid approach) shows marked improvement compared with the pure SC MPPT. The success rate (93%) is much higher than that of MCS or FPA. Thus, the advantages offered by hybridisation are worth considering.

3.3.2 Convergence time: Fig. 25 shows the average convergence time recorded by the competing algorithms. The MIC, by far, has

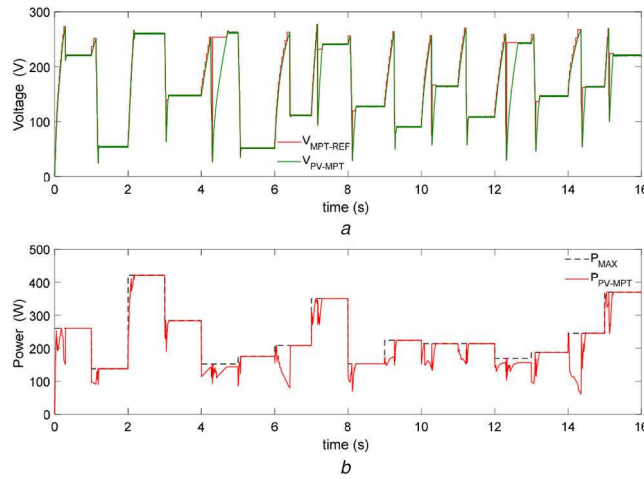


Fig. 19 Power and voltage fluctuations of the MPT algorithm [8]
(a) Voltage fluctuations, (b) Power fluctuations

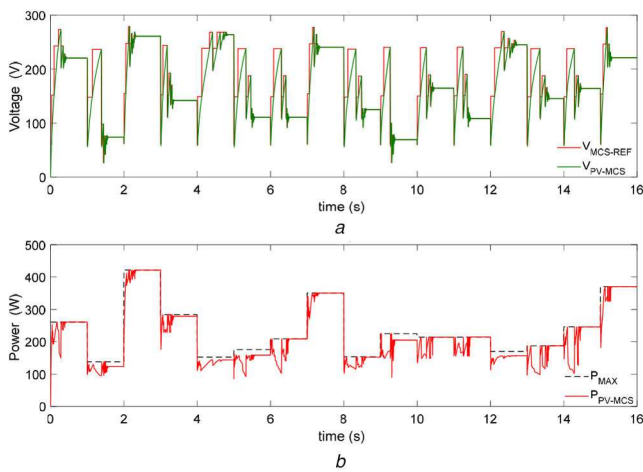


Fig. 20 Power and voltage fluctuations of the MCS algorithm [100]
(a) Voltage fluctuations, (b) Power fluctuations

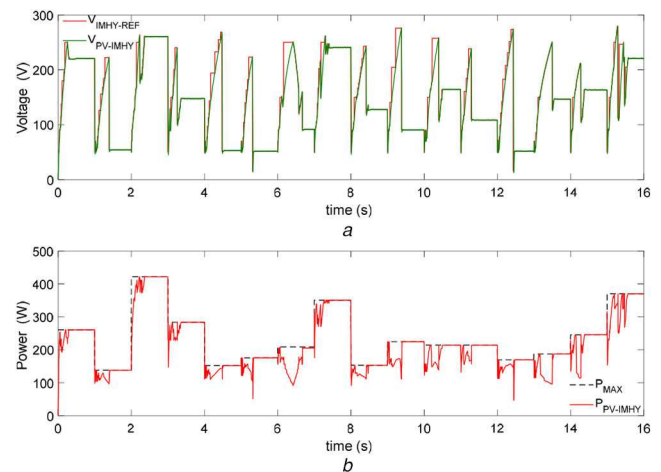


Fig. 22 Power and voltage fluctuations of the EHY algorithm [25]
(a) Voltage fluctuations, (b) Power fluctuations

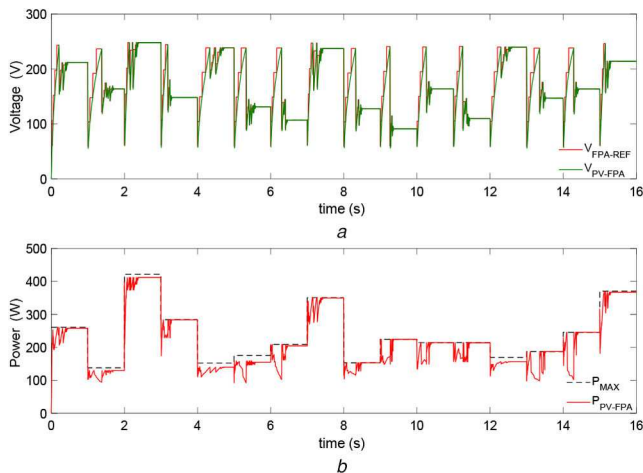


Fig. 21 Power and voltage fluctuations of the FPA algorithm [51]
(a) Voltage fluctuations, (b) Power fluctuations

the fastest convergence time (149 ms). This result is expected because it directly targets the GMPP by evaluating the $0.8 \times V_{oc}$ clusters. Once the MHP is found, the search is halted. This procedure results in rapid detection but at the expense of possible erroneous detection of GMPP. The ST, which also belongs to the $0.8 \times V_{oc}$ model, is much slower (440 ms) because it needs to visit the vicinity of all $0.8 \times V_{oc}$ before it can conclude the exact position of the GMPP. The MPT (310 ms) and VWS (275 ms), which incorporate the skipping mechanism, perform well. They are much

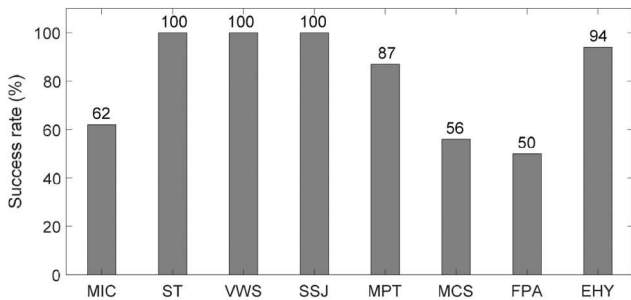
faster than SSJ (503 ms) and ST (440 ms). Notably, SSJ, which also utilises the skipping mechanism, is quite sluggish because of the utilisation of InCond in scanning the entire $P-V$ curve. The EHY is slightly faster than SSJ and ST but is much slower than MPT or VWS. Again, the hybridisation is able to reduce the tracking time due to the incorporation of SSJ into the PSO algorithm. As expected, the SC GMPPT algorithms are the slowest because of the random movements of the agents, which are manifested by the voltage fluctuations. In addition, there is a high possibility that one agent explores the space that has been scanned by other agents. This unnecessary operation is due to the lack of communication between the PSO agents, as highlighted by Kermadi *et al.* [25]. Fig. 26 shows the profile of the FPA (SC) and VWS. Clearly, the former requires more time to settle down to GMPP for every transition.

3.3.3 Voltage track length: The voltage track is defined as the path length that the operating point needs to traverse (along the $P-V$ curve voltage axis) until it converges to GMPP [8, 10]. It provides an insight into the power and voltage fluctuations that take place during the search process. Fig. 27 shows the average voltage track acquired by the competing algorithms. As can be seen from the histogram of Fig. 28, the MCS (which is a SC algorithm) exhibits longer voltage tracks compared with the VWS. This result is expected due to the random and haphazard movements of the search agents, which results in severe voltage fluctuations. This finding can be confirmed by observing the voltage (versus time) profile of the MCS, depicted in Fig. 28b. The voltage track of the VWS is $\sim 50\%$ lower, as shown in Fig. 28a.

Table 4 Detailed simulation results of all evaluation criteria for the computing GMPPT methods

Profile		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
correct tracking of the GMPP	MIC	0	1	0	1	1	1	1	0	0	1	0	1	1	0	1	1
	ST	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	VWS	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	SSJ	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	MPT	1	1	1	1	0	1	1	1	1	1	1	1	0	1	1	1
	MCS	1	0	1	0	0	0	1	1	0	0	1	1	0	1	1	1
	FPA	0	0	0	1	0	0	0	1	1	1	1	1	0	1	1	0
	EHY	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1
convergence time, ms	MIC	126	48	260	203	88	66	155	185	112	104	79	253	71	71	266	303
	ST	482	280	447	297	392	534	540	532	262	376	580	375	422	663	508	359
	VWS	272	393	184	191	455	277	239	182	334	242	217	223	380	359	264	190
	SSJ	450	439	537	477	525	395	636	419	716	384	600	472	455	532	421	591
	MPT	298	201	171	89	690	75	430	290	205	301	396	277	638	207	477	228
	MCS	432	571	325	364	823	571	514	347	612	400	467	460	723	606	484	316
	FPA	461	715	390	302	731	598	470	436	506	363	409	454	593	529	426	310
	EHY	429	384	410	377	470	358	399	396	444	402	547	455	404	423	384	628
voltage track, V	MIC	122	85	236	201	95	91	163	155	123	126	101	262	91	93	205	261
	ST	409	225	465	360	229	232	283	450	293	287	343	292	237	315	334	422
	VWS	273	573	233	452	364	582	272	239	484	373	261	387	501	290	315	317
	SSJ	292	383	298	350	470	398	438	293	427	322	362	379	452	339	214	336
	MPT	243	287	211	271	558	242	211	483	331	376	468	361	589	316	438	393
	MCS	493	794	512	784	542	789	636	511	778	701	543	701	520	733	612	544
	FPA	578	672	755	641	518	785	602	686	675	552	484	688	437	634	525	587
	EHY	263	231	397	531	155	276	248	262	502	133	407	445	119	251	411	844
tracking efficiency, %	MIC	91	98	85	93	98	98	95	89	89	96	71	91	97	87	89	86
	ST	84	89	85	89	87	76	69	81	90	81	76	84	88	66	62	86
	VWS	90	86	94	95	86	90	89	97	88	88	90	93	90	77	82	94
	SSJ	79	83	73	76	82	82	70	80	74	77	73	78	84	68	75	66
	MPT	88	89	95	97	83	96	69	94	90	84	83	89	85	87	56	95
	MCS	92	85	93	97	89	73	66	97	86	81	81	89	89	64	54	94
	FPA	88	82	92	93	84	79	78	92	86	84	87	86	88	71	76	92
	EHY	90	88	91	91	86	85	81	94	84	80	78	85	89	80	81	84

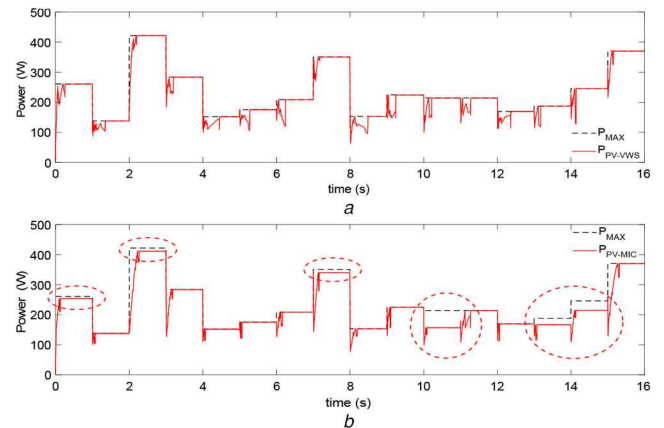
Bold values indicate the significance of shades is to indicate that the harvested power is not maximal.

**Fig. 23** Success rate (% of convergence to GMPP) for all GMPPTs

3.3.4 Transient efficiency: The transient efficiency (η) is calculated using the following expression:

$$\eta = \frac{\int_{T_i}^{T_f} (P(t)) dt}{\int_{T_i}^{T_f} (P_{\max}(t)) dt} \times 100 \quad (14)$$

where $P_{\max}$ represents the maximum theoretical power available during that time interval. T_i and T_f are the lower and upper time limits of the efficiency computation, respectively. The duration between these two limits is set as 500 ms. Fig. 29 shows the average transient efficiencies for the competing GMPPT algorithms. The lowest is recorded by SSJ (76%), followed by ST (80%). The low efficiency is closely related to the voltage track; generally, a shorter voltage track results in better efficiency. Hence, the highest efficiency is achieved by MIC. Similarly, MCS, FPA,

**Fig. 24** Power profile for the 16 shading patterns to show the convergence to GMPP for (a) VWS, (b) MIC

EHY, and MPT have high efficiency (between 84 and 86%), while the SC methods have lower efficiency.

3.4 Benchmarking

Fig. 30 shows a graphical representation of the benchmark of the competing GMPPT methods with respect to the four evaluation criteria. The contours, which are normalised to the scale of 1 (minimum) to 5 (maximum), suggest the weakness and strengths of the methods when subjected to the scenario of the 16 shading patterns. However, ranking them would be misleading because no

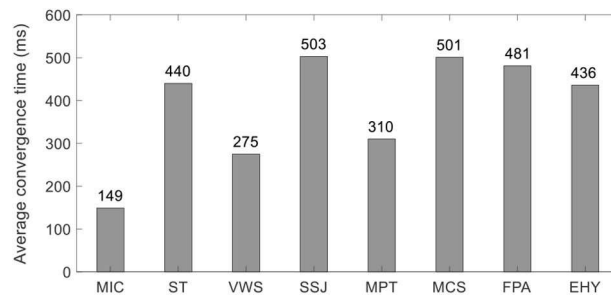


Fig. 25 Average convergence time

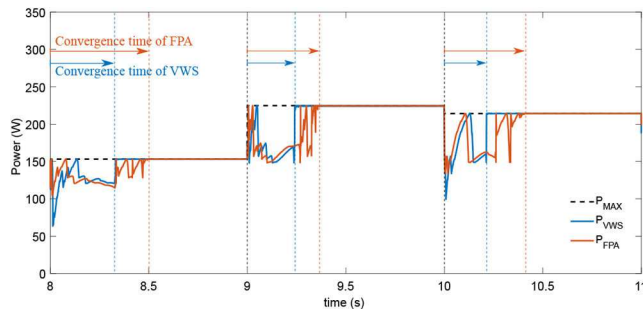


Fig. 26 Comparison of the convergence time for VWS and FPA

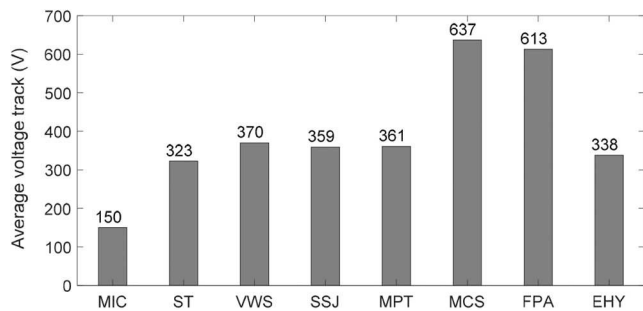


Fig. 27 Average voltage track length

single method commands superiority over others in all evaluation aspects. Despite the inconclusiveness of the results, the following general observations can be used as guidelines to select the GMPPT for relevant applications:

- MIC (Fig. 30a) outperforms other GMPPTs in terms of voltage track, speed, and transient efficiency. However, the success rate is comparatively low.
- ST (Fig. 30b) and SSJ (Fig. 30d) show excellent tracking success with an optimised voltage track. However, SSJ is slow to converge, which significantly affects its tracking efficiency.
- The GMPPTs that utilise the skipping mechanism, i.e. VWS (Fig. 30c) and MPT (Fig. 30e), exhibit the best performance in terms of tracking success, speed, and voltage track.
- The SC-based GMPPTs, i.e. MCS (Fig. 30f) and FPA (Fig. 30g), have the lowest performance for all criteria.
- The hybrid approach mitigates the limitations of SC GMPPT as evidenced by the improved performance of EHY (Fig. 30h).

4 Conclusion

This paper reviews the state-of-the-art GMPPT for PV systems under partial shading conditions. GMPPTs are classified into four main categories: modified conventional method (which is mainly based on the $0.8 \times V_{oc}$ model), skipping mechanism, SC, and hybrid. Then, comprehensive comparative and assessment analyses between selected algorithms are carried out. Benchmarking is performed by subjecting the algorithms to the same simulation platform and operating conditions. The performance is evaluated based on four assessment criteria: success rate (convergence to GMPP), convergence time, voltage track length, and transient

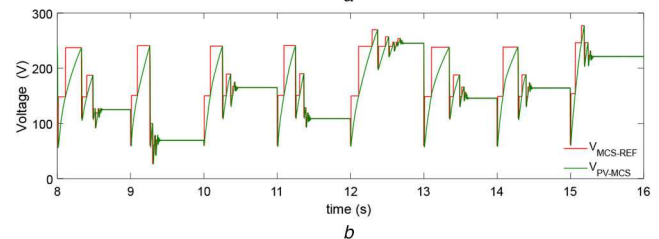
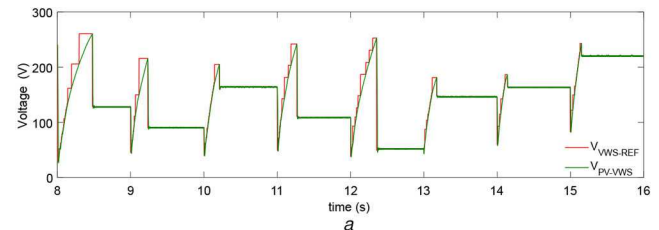


Fig. 28 Voltage track length
(a) MCS, (b) VWS

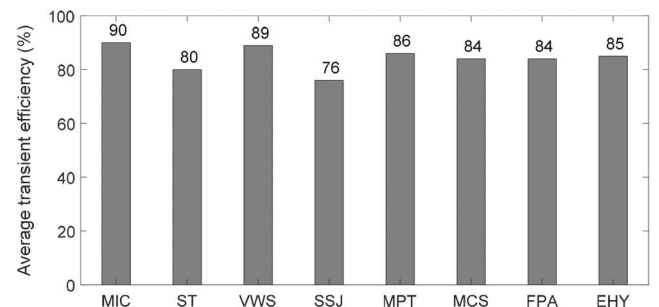


Fig. 29 Average transient efficiency

efficiency. The GMPPT based on skipping mechanisms is the most promising, providing the best results in terms of tracking speed and voltage fluctuation during transitions and successfully tracking the GMPP under all shading conditions.

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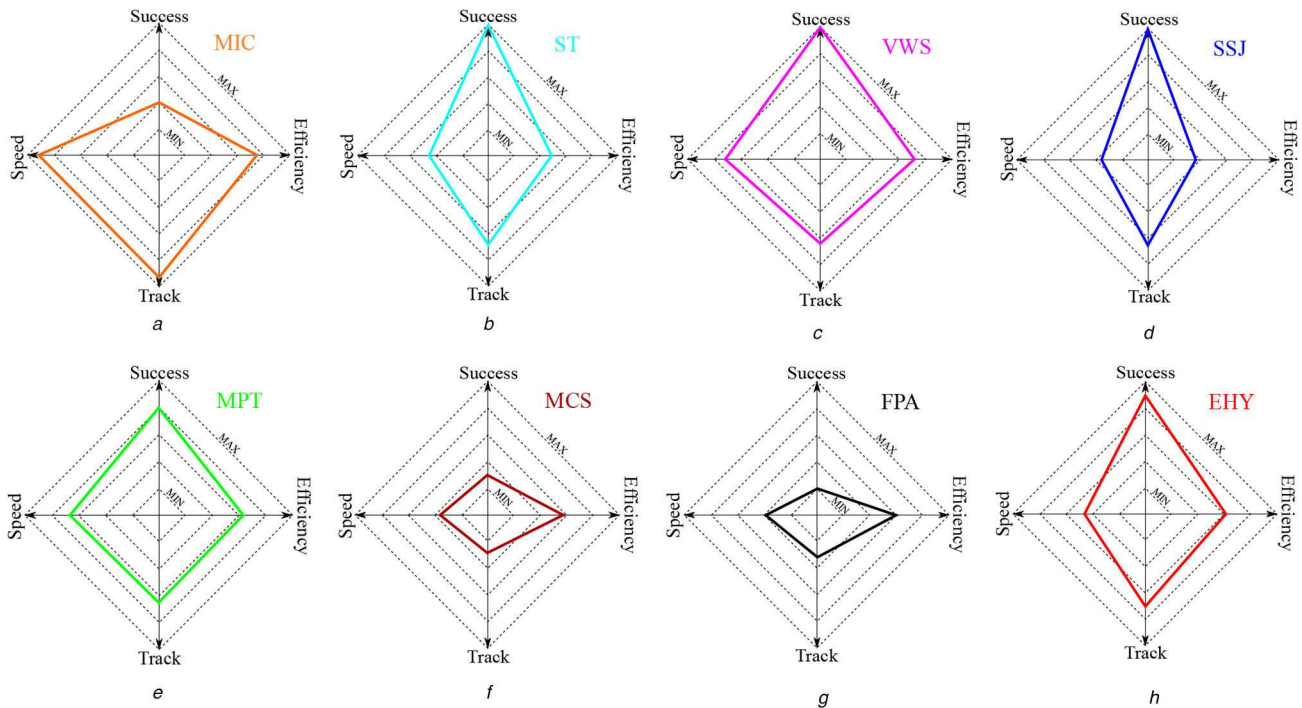


Fig. 30 Strength of the compared GMPPT algorithms with respect to each evaluation criterion
(a) MIC, (b) ST, (c) VWS, (d) SSI, (e) MPT, (f) MCS, (g) FPA, (h) EHY

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